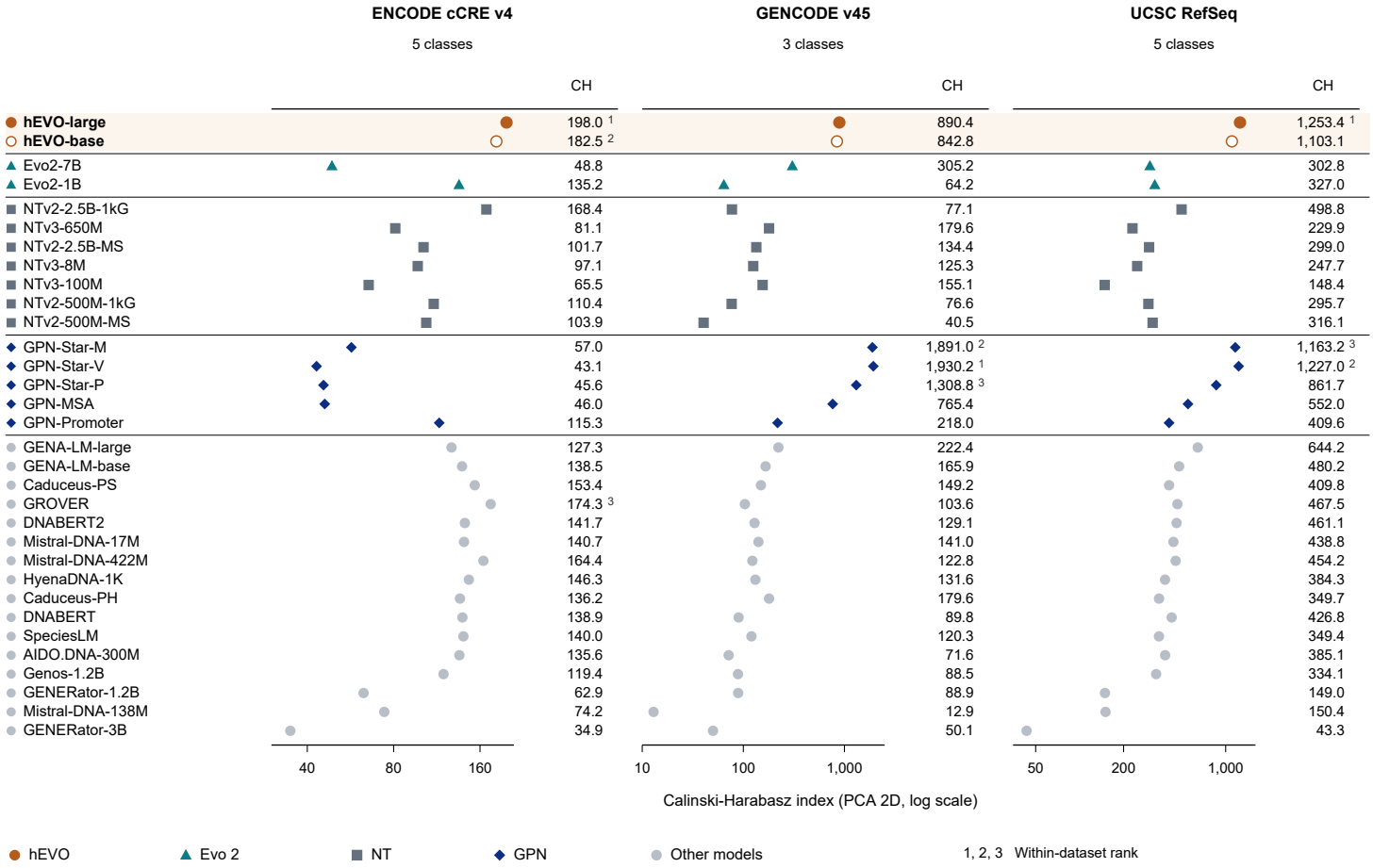

ENCODE cCRE v4
5 classes
GENCODE v45
3 classes
UCSC RefSeq
5 classes
CH
hEVO-large 198.0 1 890.4 1,253.4 1
hEVO-base 182.5 2 842.8 1,103.1
Evo2-7B 48.8 305.2 302.8
Evo2-1B 135.2 64.2 327.0
NTv2-2.5B-1kG 168.4 77.1 498.8
NTv3-650M 81.1 179.6 229.9
NTv2-2.5B-MS 101.7 134.4 299.0
NTv3-8M 97.1 125.3 247.7
NTv3-100M 65.5 155.1 148.4
NTv2-500M-1kG 110.4 76.6 295.7
NTv2-500M-MS 103.9 40.5 316.1
GPN-Star-M 57.0 1,891.0 2 1,163.2 3
GPN-Star-V 43.1 1,930.2 1 1,227.0 2
GPN-Star-P 45.6 1,308.8 3 861.7
GPN-MSA 46.0 765.4 552.0
GPN-Promoter 115.3 218.0 409.6
GENA-LM-large 127.3 222.4 644.2
GENA-LM-base 138.5 165.9 480.2
Caduceus-PS 153.4 149.2 409.8
GROVER 174.3 3 103.6 467.5
DNABERT2 141.7 129.1 461.1
Mistral-DNA-17M 140.7 141.0 438.8
Mistral-DNA-422M 164.4 122.8 454.2
HyenaDNA-1K 146.3 131.6 384.3
Caduceus-PH 136.2 179.6 349.7
DNABERT 138.9 89.8 426.8
SpeciesLM 140.0 120.3 349.4
AIDO.DNA-300M 135.6 71.6 385.1
Genos-1.2B 119.4 88.5 334.1
GENERator-1.2B 62.9 88.9 149.0
Mistral-DNA-138M 74.2 12.9 150.4
GENERator-3B 34.9 50.1 43.3
40
80
160
10
100
1,000
50
200
1,000
Calinski-Harabasz index (PCA 2D, log scale)
hEVO
Evo 2
NT
GPN
Other models
1, 2, 3 Within-dataset rank